* Pythagoras Theorem

Let $S = \{u_1, u_2, \ldots, u_n\}$ is an orthogonal set then prove that

$$\|u_1 + u_2 + \ldots + u_n\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_n\|^2$$

Proof:- $\|u_1 + u_2 + \ldots + u_n\|^2 = \langle u_1 + u_2 + \ldots + u_n, u_1 + u_2 + \ldots + u_n \rangle$

$$= \langle u_1, u_1 \rangle + \langle u_1, u_2 \rangle + \ldots + \langle u_1, u_n \rangle$$
$$+ \langle u_2, u_1 \rangle + \langle u_2, u_2 \rangle + \ldots + \langle u_2, u_n \rangle$$
$$+ \ldots$$
$$+ \langle u_n, u_1 \rangle + \langle u_n, u_2 \rangle + \ldots + \langle u_n, u_n \rangle$$
$$= \langle u_1, u_1 \rangle + \langle u_2, u_2 \rangle + \ldots + \langle u_n, u_n \rangle$$

$[\because \langle u_i, u_j \rangle = 0$ for $i \neq j]$

$$= \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_n\|^2$$

12.03.2020

Cauchy-Schwartz Inequality :-

For u and v in a real ~~vector~~ inner product space V,

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

Proof:- $\langle tu + v, tu + v \rangle$; $t \in \mathbb{R}$

$$= \langle tu, tu \rangle + \langle tu, v \rangle + \langle v, tu \rangle + \langle v, v \rangle$$
$$= \langle u, u \rangle t^2 + t\langle u, v \rangle + t\langle v, u \rangle + \langle v, v \rangle$$
$$= \|u\|^2 t^2 + 2t\langle u, v \rangle + \|v\|^2$$

P Kalika Maths